

From Statcast Data to Actionable MLB Scouting Intelligence

Aaron Perez Sales / DATA 510 Data Science Capstone · Summer 2026

ABSTRACT

This project develops an automated baseball scouting and matchup analysis system that transforms MLB pitch-level data into interactive decision-support reports.

Data is collected from the MLB Stats API, validated, and stored in a PostgreSQL database hosted on Railway before being analyzed using Python. The final output is an interactive HTML report that provides coaches and analysts with actionable pregame insights.

INTRODUCTION & MOTIVATION

Modern baseball organizations collect millions of pitch observations, making manual pregame analysis difficult. This automated decision-support system leverages centralized PostgreSQL and python analytics to generate coach-ready matchup insights.

Research Question

How can pitch level MLB data be transformed into an automated, reproducible, and interpretable decision support system for opponent scouting and matchup preparation?

Primary Stakeholders

Coaches · Professional Scouts · Front Office Analysts · Player Development

BACKGROUND & RELATED WORK

Related Work & Foundations

- Sabermetrics and advanced baseball analytics models
- MLB Statcast tracking technology and API pipelines
- Sports decision-support and visualization interfaces
- PostgreSQL data warehouse design for analytical workloads

Key References

- **Lewis (2003)** *Moneyball*
- **Tango et al. (2007)** *The Book: Playing the Percentages in Baseball*
- **Alamar (2013)** *Sports Analytics*

DATA, ENGINEERING & ETHICS

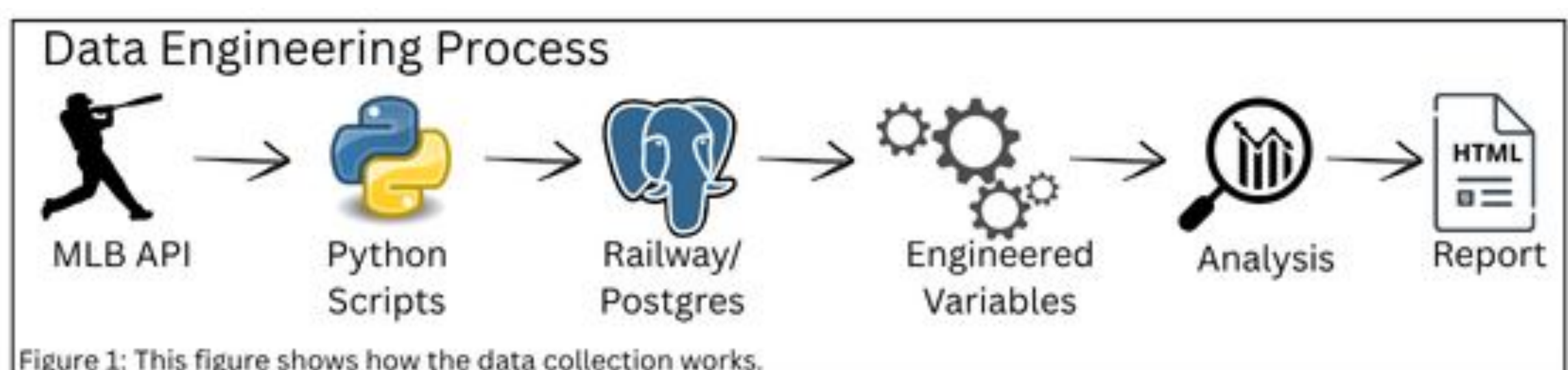
Data Architecture & Pipeline

MLB Stats API Endpoints:

- **Schedule:** Maps game schedules, dates, and keys
- **Live Feed:** Extracts pitch-by-pitch real-time events
- **Player/Team:** Demographic details, throwing/batting hands of current & historical player rosters

Storage & Infrastructure:

- **PostgreSQL:** Standard relational warehouse for normalized player, team, game, and pitch metrics.
- **Railway:** Cloud-hosted DB instance with automated pipelines.



Ethics & Data Security

- Public baseball performance data only
- No sensitive personal information collected
- Small-sample limitations acknowledged
- Transparent handling of missing data
- Reproducible analytical workflow

METHODS & EVALUATION

Methods

Data Collection & Engineering

- MLB Stats API pitch-by-pitch data collection & verification.
- Calculated platoon splits, pitch usage, and strike-zone tendencies.
- Normalized PostgreSQL database hosted on Railway.

Report Generation

- Interactive HTML reports built with Python
- Interactive strike-zone heatmaps and release-point graphics.

Evaluation & Validation

Evaluating scouting workflow execution over model accuracy.

- Verified left/right matchup calculations against PostgreSQL queries.
- Confirmed heatmaps and release-points reflect actual pitch locations.

RESULTS & VISUALIZATIONS

Batter Attack Plan

What this hitter should attack

Purpose: Match this hitter's damage profile only to pitches the opposing pitcher actually throws.

Pitches to attack

- **Four-Seam Fastball:** 60.0% hard-hit rate over 360 pitches
- **Curveball:** 55.6% hard-hit rate over 76 pitches
- **Sinker:** 44.8% hard-hit rate over 313 pitches

Locations to attack

- **Upper-left third:** 75.0% hard-hit rate over 48 pitches
- **Heart of zone:** 67.5% hard-hit rate over 115 pitches
- **Lower-right third:** 66.7% hard-hit rate over 51 pitches

Pitches to avoid

- Slider
- Splitter
- Curveball

Fig 2: This figure shows what pitches our batter should attack

Pitcher's Pitch Location

Four-Seam Fastball

31.6% usage

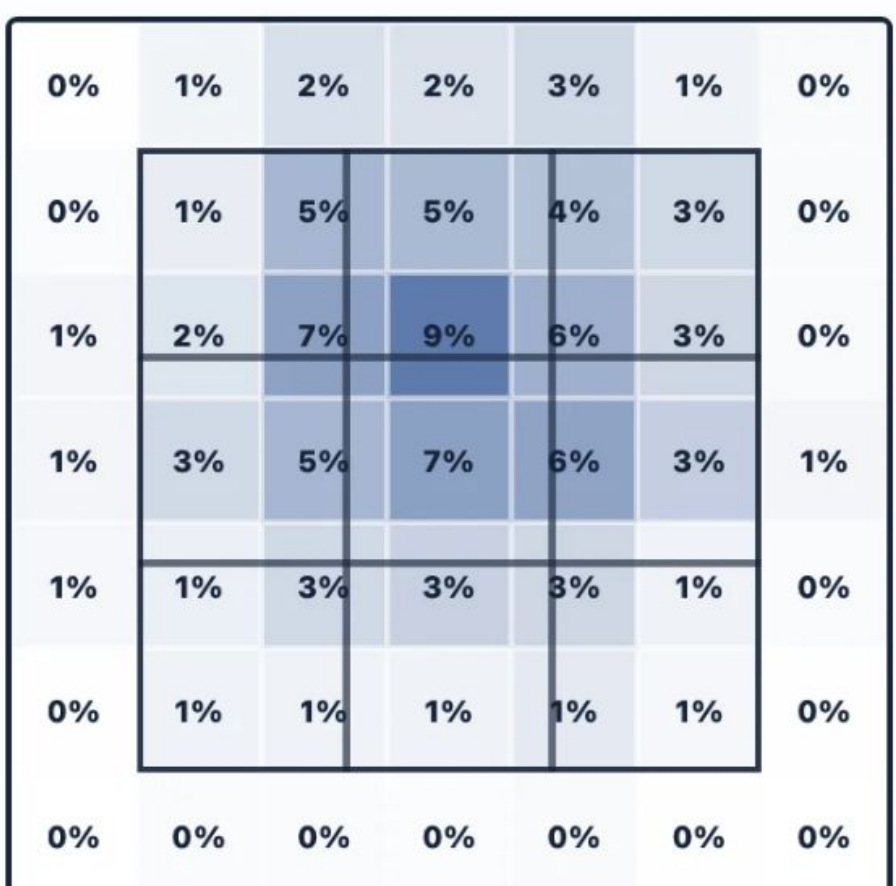


Fig 3: This figure shows an example of pitcher's location tendency by pitch

5 Game Rolling Trends

Whiff / swing

45.8%



Fig 4: This figure shows an example of the 5-game rolling averages looks likes on the report

Pitcher's Pitch Tendency

0-0

Pitcher count tendency

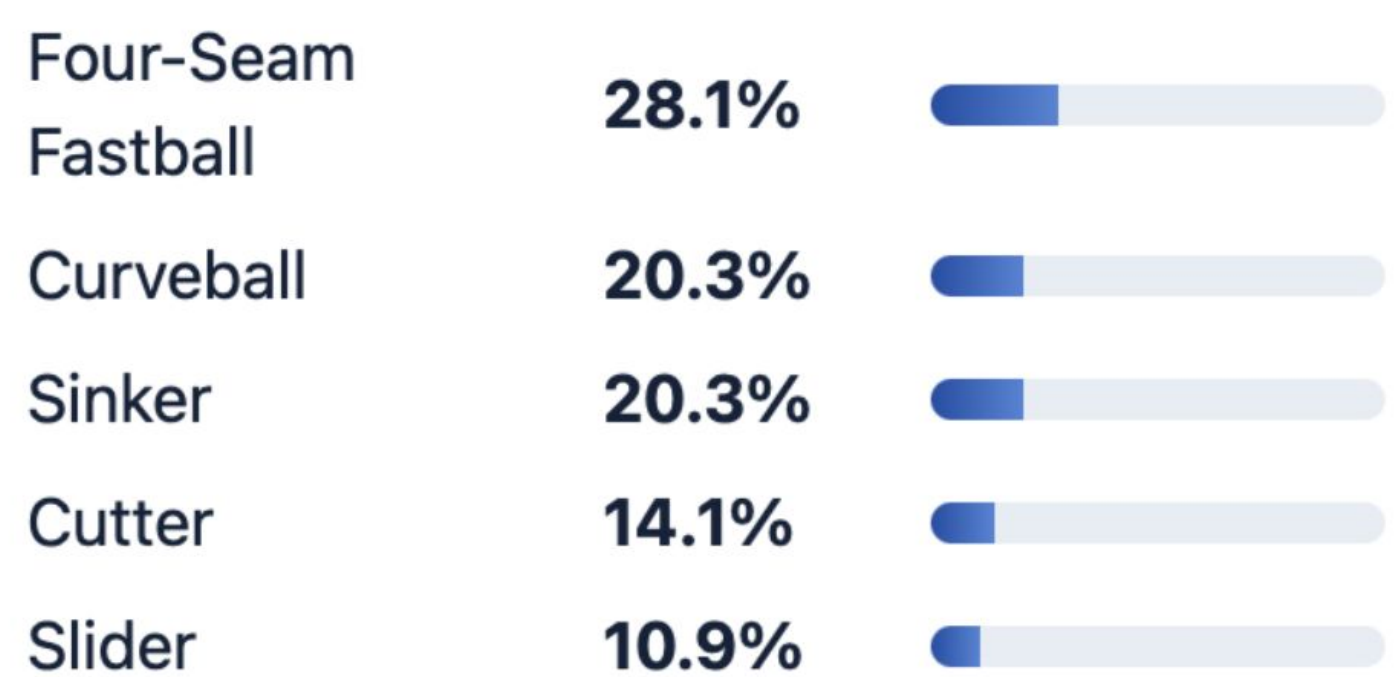


Fig 5: This figure shows an example of the report section that focuses on count and pitch selection.

Batter Hit Tendency

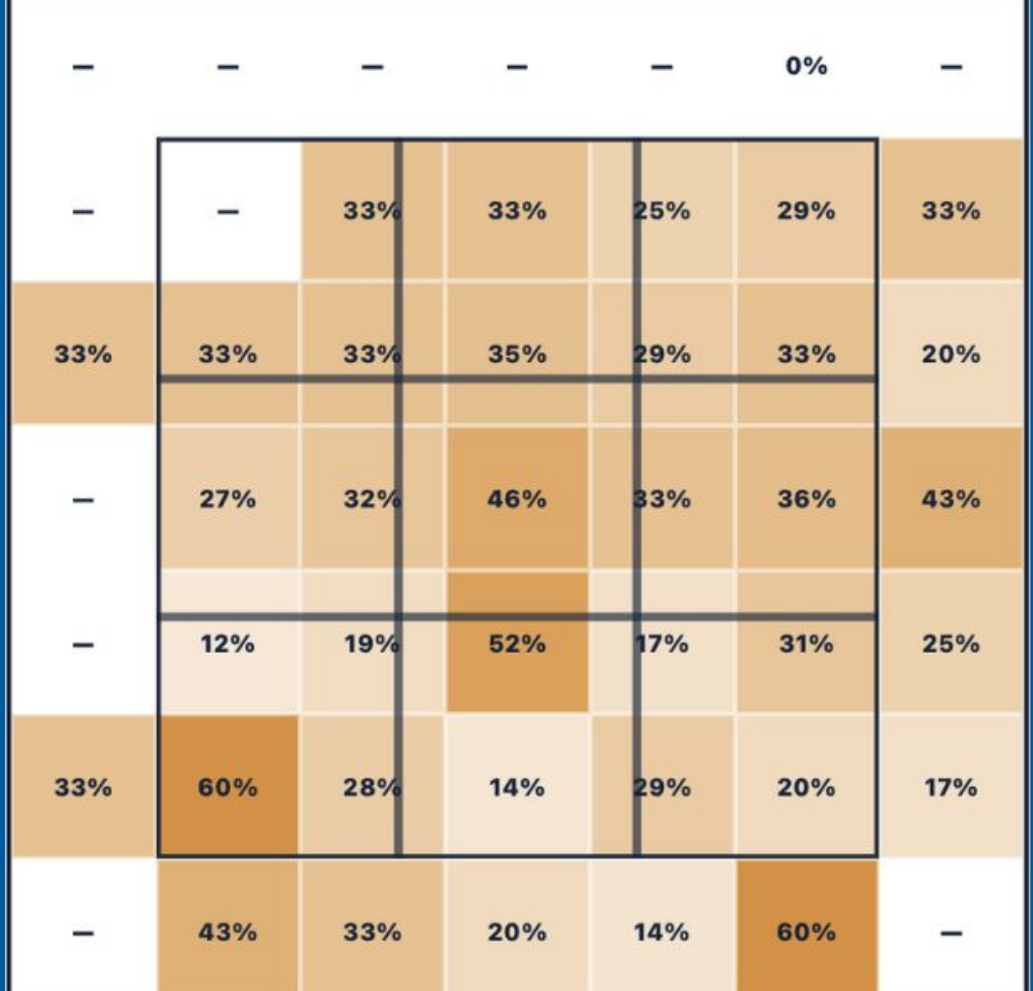


Fig 6: This example from the report shows where a certain batter hit percentages from an area in the strike zone

DISCUSSION & CONCLUSIONS

Current Limitations

- Small sample sizes for newly promoted players
- Historical performance splits may not predict sudden mechanical changes
- Excludes immediate in-game environmental and bullpen adjustments

Conclusions & Recommendations

- **Integration:** Successfully integrated data engineering, analytics, and visualization into a unified scouting platform.
- **Value:** Demonstrated the value of matchup-specific baseball analytics.
- **Accessibility:** Produced interactive reports that improve accessibility for coaches and analysts.
- **Scalability:** Established a scalable framework for future predictive modeling.

Reproducibility & References

Lewis, M. (2003). *Moneyball* · Tango, T., Lichtman, M., & Dolphin, A. (2007). *The Book* · Alamar, B. C. (2013). *Sports Analytics* · Kleppmann, M. (2017). *Designing Data-Intensive Applications* · MLB Statcast Docs